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Course : Machine Learning

Poject : Spam Email Detector

Batch : CS-FA22 8th SEM

Department : C&T

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Islamabad**

Project Name : Spam Email Detector

Problem : Classify as emails as Spam or Not-spam.

Code :-

Cell 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import nltk
import re
import string
from wordcloud import WordCloud
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    roc_curve,
    auc
)
import pickle
```

Cell 2: Download Resources

```
nltk.download('stopwords')
```

Cell 3: Load Dataset

```
df = pd.read_csv('spam.csv', encoding='latin-1')
df.head()
```

Output :-

...	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN

Cell 4: Remove Extra Columns

```
df = df[['v1','v2']]
df.columns = ['label','message']
df.head()
```

Output :-

...	label	message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

Cell 5: Dataset Information

```
df.info()
print(df.shape)
```

Output :-

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   label       5572 non-null   object
1   message     5572 non-null   object
dtypes: object(2)
memory usage: 87.2+ KB
(5572, 2)
```

Cell 6: Missing Values

```
print(df.isnull().sum())
```

Cell 7: Check Class Distribution

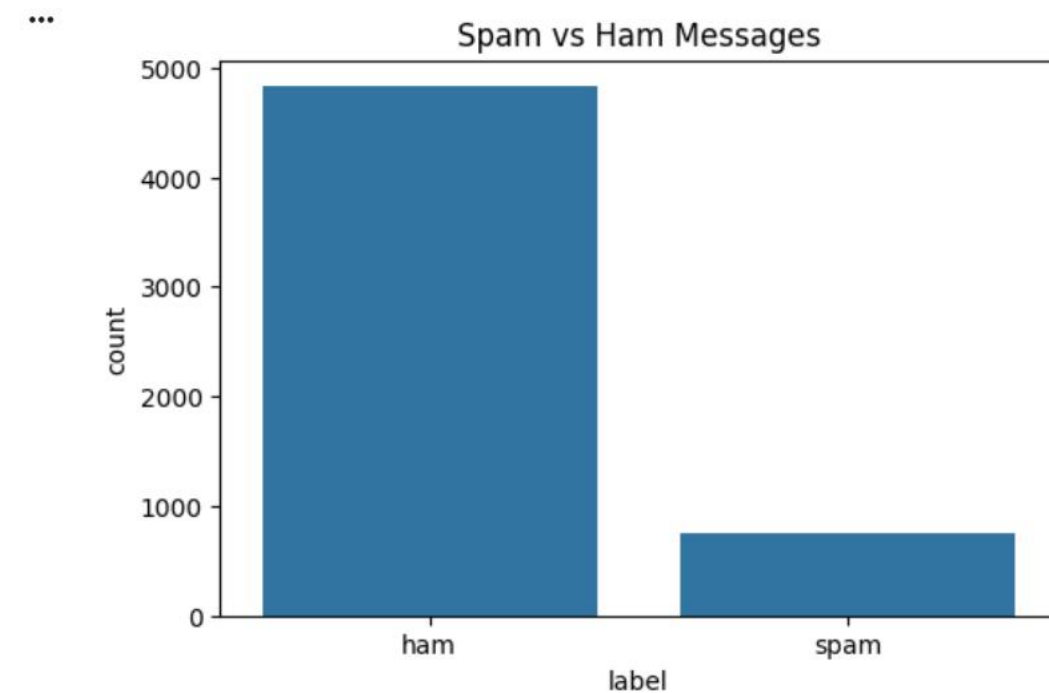
```
df['label'].value_counts()
```

Output :-

```
...          count
label
ham          4825
spam          747
dtype: int64
```

Cell 8: Visualize Spam vs Ham

```
plt.figure(figsize=(6,4))
sns.countplot(x='label',data=df)
plt.title("Spam vs Ham Messages")
plt.show()
```



Cell 9: Message Length Feature

```
df['length'] = df['message'].apply(len)
df.head()
```

Output :-

...	label	message	length
0	ham	Go until jurong point, crazy.. Available only ...	111
1	ham	Ok lar... Joking wif u oni...	29
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	155
3	ham	U dun say so early hor... U c already then say...	49
4	ham	Nah I don't think he goes to usf, he lives aro...	61

Cell 10: Length Distribution

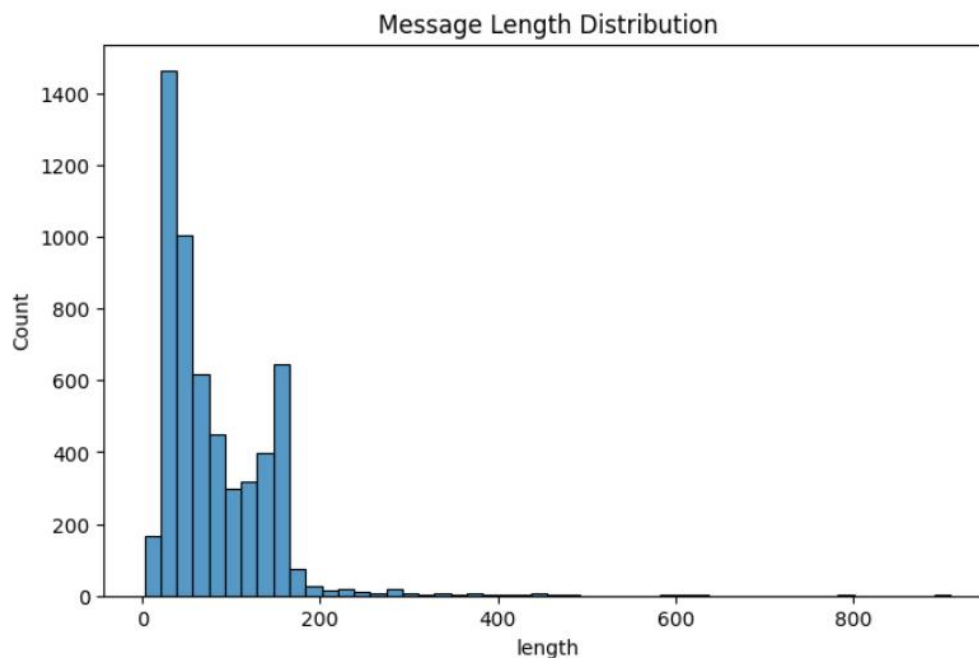
```
plt.figure(figsize=(8,5))

sns.histplot(df['length'], bins=50)

plt.title("Message Length Distribution")

plt.show()
```

Output :-



Cell 11: Text Cleaning Function

```
stop_words = set(stopwords.words('english'))
```

```
ps = PorterStemmer()
```

```
def clean_text(text):
```

```
    text = text.lower()
```

```
    text = re.sub(r'http\S+', '', text)
```

```
    text = re.sub(r'\d+', '', text)
```

```
    text = text.translate(
        str.maketrans("", string.punctuation)
    )
```

```
    words = text.split()
```

```
    words = [
        ps.stem(word)
        for word in words
        if word not in stop_words
    ]
```

```
    return " ".join(words)
```

Cell 12: Apply Cleaning

```
df['message'] = df['message'].apply(clean_text)
```

```
df.head()
```

Output :-

	label	message	length
0	ham	go jurong point crazi avail bugi n great world...	111
1	ham	ok lar joke wif u oni	29
2	spam	free entri wkli comp win fa cup final tkt st m...	155
3	ham	u dun say earli hor u c already say	49
4	ham	nah dont think goe usf live around though	61

Cell 13: Label Encoding

```
df['label'] = df['label'].map({
    'ham':0,
    'spam':1
})
```

Cell 14: Word Cloud For Spam

```
spam_words = " ".join(
    df[df['label']==1]['message']
)
```

```
wordcloud = WordCloud(
    width=800,
    height=400,
    background_color='white'
).generate(spam_words)
```

```
plt.figure(figsize=(12,6))
```

```
plt.imshow(wordcloud)
```

```
plt.axis('off')
```

```
plt.title("Spam Word Cloud")
```

```
plt.show()
```

Output :-



```
ham_words = " ".join(
    df[df['label']==0]['message']
)
```

```
plt.figure(figsize=(12,6))
```

```
plt.axis('off')
```

```
plt.show()
```

[illegible]

```
X = df['message']
```

```
y = df['label']
```


Cell 17: TF-IDF Vectorization

```
vectorizer = TfidfVectorizer(  
    max_features=5000  
)
```

```
X = vectorizer.fit_transform(X)
```

Cell 18: Split Dataset

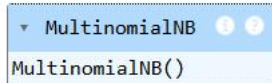
```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

Cell 19: Train Naive Bayes

```
nb_model = MultinomialNB()
```

```
nb_model.fit(  
    X_train,  
    y_train  
)
```

Output :-

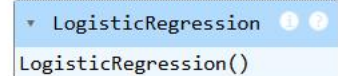


A screenshot of a Jupyter Notebook cell output. It shows a dropdown menu with 'MultinomialNB' selected, followed by a light blue box containing the text 'MultinomialNB()'.

Cell 20: Train Logistic Regression

```
lr_model = LogisticRegression()  
lr_model.fit(  
    X_train,  
    y_train  
)
```

Output :-



A screenshot of a Jupyter Notebook cell output. It shows a dropdown menu with 'LogisticRegression' selected, followed by a light blue box containing the text 'LogisticRegression()'.

Cell 21: Compare Models

```
nb_pred = nb_model.predict(X_test)
```

```
lr_pred = lr_model.predict(X_test)
```

```
print(
    "Naive Bayes Accuracy:",
    accuracy_score(y_test,nb_pred)
)
```

```
print(
    "Logistic Regression Accuracy:",
    accuracy_score(y_test,lr_pred)
)
```

Output :-

```
... Naive Bayes Accuracy: 0.9713004484304932
    Logistic Regression Accuracy: 0.9488789237668162
```

Cell 22: Classification Report

```
print(
    classification_report(
        y_test,
        nb_pred
    )
)
```

Output :-

...		precision	recall	f1-score	support
	0	0.97	1.00	0.98	965
	1	1.00	0.79	0.88	150
	accuracy			0.97	1115
	macro avg	0.98	0.89	0.93	1115
	weighted avg	0.97	0.97	0.97	1115

Cell 23: Confusion Matrix

```
cm = confusion_matrix(
    y_test,
    nb_pred
)
```

```
plt.figure(figsize=(6,5))
```

```

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.xlabel("Predicted")

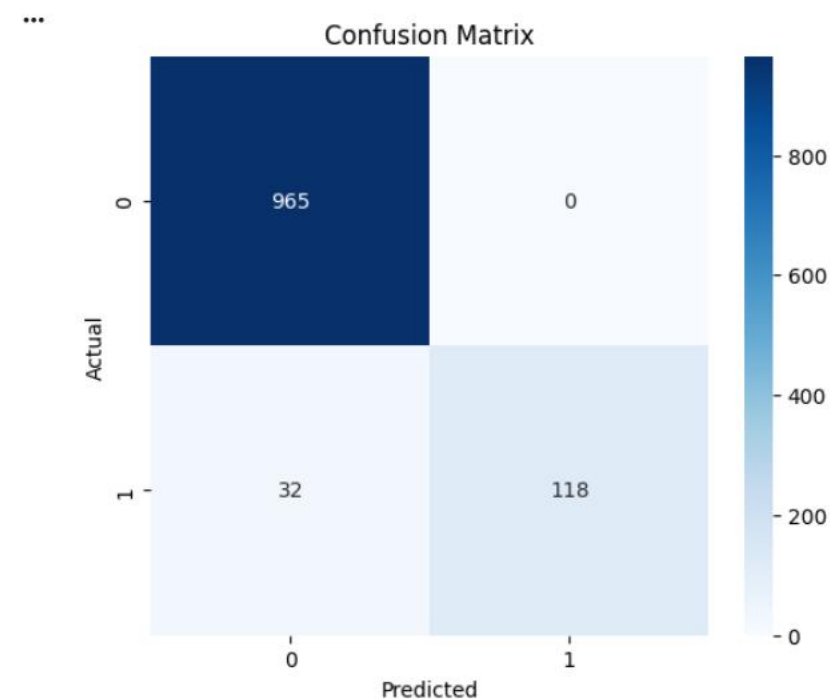
plt.ylabel("Actual")

plt.title("Confusion Matrix")

plt.show()

```

Output :-



Cell 24: ROC Curve

```

probs = nb_model.predict_proba(
    X_test
)[: ,1]

fpr,tpr,_ = roc_curve(
    y_test,

```

```

    probs
)

roc_auc = auc(
    fpr,
    tpr
)

plt.figure(figsize=(7,5))

plt.plot(
    fpr,
    tpr,
    label=f"AUC = {roc_auc:.2f}"
)

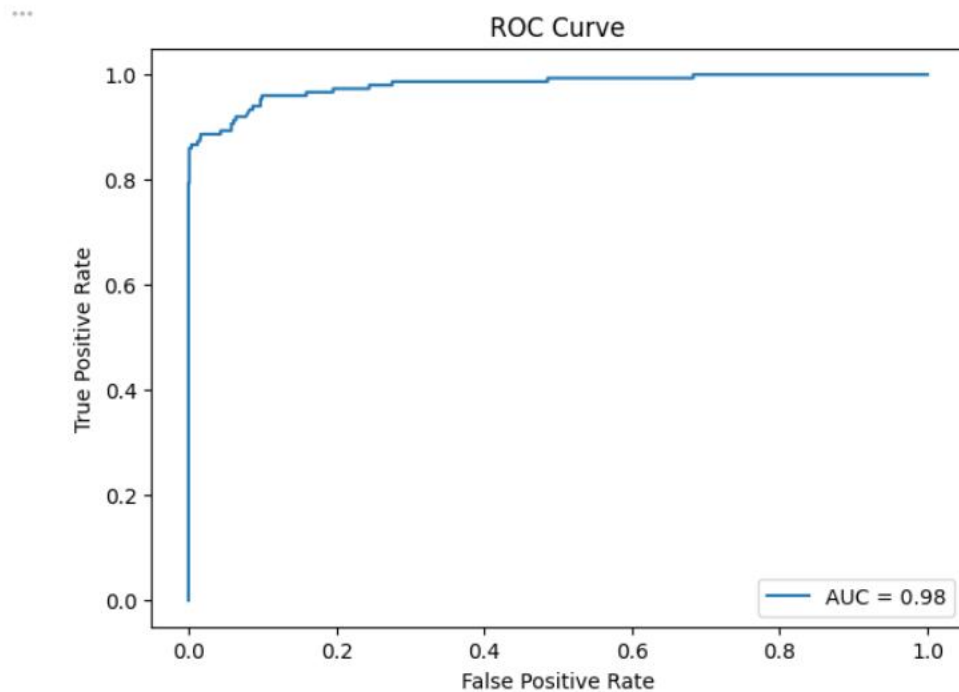
plt.legend()
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")

plt.title("ROC Curve")

plt.show()

```

Output :-



Cell 25: Interactive Testing

while True:

```
msg = input(
    "Enter Message: "
)

msg = clean_text(msg)

msg_vector = vectorizer.transform([msg])

result = nb_model.predict(
    msg_vector
)

if result[0] == 1:
    print("🚨 SPAM MESSAGE")
else:
    print("✅ NOT SPAM")

again = input(
    "Check another? (y/n): "
)

if again.lower() != 'y':
    break
```

Output :-

```
*** Enter Message: URGENT! Claim your $5000 cash prize today.
🚨 SPAM MESSAGE
Check another? (y/n): y
Enter Message: FREE entry into our competition. Text WIN now!
🚨 SPAM MESSAGE
Check another? (y/n): y
Enter Message: My self Saif ur rehman.
✅ NOT SPAM
Check another? (y/n): y
Enter Message: I'm live in islamabad and i'm styding in iqra university.
✅ NOT SPAM
Check another? (y/n): n
```

Cell 26: Save Model

```
pickle.dump(
    nb_model,
    open(
        "spam_model.pkl",
```

```
        "wb"  
    )  
)  
  
pickle.dump(  
    vectorizer,  
    open(  
        "vectorizer.pkl",  
        "wb"  
    )  
)  
  
print(  
    "Model Saved Successfully"  
)
```